

Module 22: Real-World Power BI Projects & Portfolio Development



Overview

Learning Power BI concepts is important, but employers hire candidates based on their ability to solve business problems.

This module focuses on building real-world Power BI projects, documenting them professionally, creating a portfolio, showcasing skills on GitHub, and preparing for interviews.

By the end of this module, you will know how to convert raw business requirements into complete analytical solutions and build a portfolio that demonstrates practical experience.



Learning Objectives

After completing this module, you will be able to:

- Execute end-to-end Power BI projects
- Gather business requirements
- Identify KPIs
- Design dashboard solutions
- Build portfolio-quality projects
- Document projects professionally
- Create GitHub repositories
- Present projects to recruiters
- Prepare for Power BI interviews



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1 Why Projects Matter

What Recruiters Look For

Recruiters care less about certifications and more about evidence of practical skills.

Projects Demonstrate

Problem Solving

Data Analysis

Dashboard Design

DAX Skills

Business Understanding

Hiring Perspective

Skills + Projects
>
Skills Alone

2 End-to-End Project Lifecycle

Typical Workflow

```
graph TD; A[Business Problem] --> B[Requirement Gathering]; B --> C[Data Collection]; C --> D[Data Cleaning]; D --> E[Data Modeling]; E --> F[DAX Development]; F --> G[Dashboard Design]; G --> H[Testing]; H --> I[Deployment]; I --> J[Documentation];
```

3 Business Requirement Gathering

First Step in Every Project

Understand the business problem.

Questions to Ask

What decision needs to be made?

Who are the stakeholders?

What KPIs are important?

What data is available?

What actions should users take?

Example

Business Problem:

Sales performance is declining.

Requirement

Identify:

- Revenue trends
- Product performance
- Regional performance



KPI Identification

What is a KPI?

Key Performance Indicator.

Sales KPIs

- Revenue
- Profit
- Margin %
- Orders
- Customers

HR KPIs

- Attrition Rate
- Employee Count
- Retention Rate

Finance KPIs

- Revenue
- Operating Cost
- Net Profit

Marketing KPIs

- Conversion Rate
- Customer Acquisition Cost

- Campaign ROI

5 Project Planning

Before opening Power BI:

Define Scope

What should be included?

Define Deliverables

Examples:

- Dashboard
- Report
- Documentation

Define Timeline

Example:

```
Week 1 → Data Preparation
Week 2 → Dashboard Development
Week 3 → Documentation
```

6 Dataset Selection

Sources

Kaggle

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Government Data

Examples:

- Open Data Portals

Company Data

Internal business systems.

Characteristics of Good Datasets

Sufficient Size

Business Relevance

Multiple Dimensions

Realistic Complexity

Data Cleaning Strategy

Most projects spend significant time cleaning data.

Common Tasks

Remove Duplicates

Handle Missing Values

Standardize Formats

Correct Data Types

Validate Data

Tools

- Power Query
 - SQL
 - Python
-

Data Modeling Strategy

Recommended Structure

Star Schema

```
DimCustomer
  ↓
FactSales
  ↑
DimProduct
```

Benefits

Better Performance

Simpler DAX

Scalability



DAX Development

Typical Measures

Total Revenue

Total Profit

Profit Margin %

Customer Count

Advanced Metrics

YTD Revenue

Growth %

Market Share

Ranking

10 Dashboard Design

Typical Layout

```
graph TD; KPIs --> TrendAnalysis[Trend Analysis]; TrendAnalysis --> CategoryAnalysis[Category Analysis]; CategoryAnalysis --> DetailedBreakdown[Detailed Breakdown];
```

Principles

Simplicity

Consistency

Business Focus

User Experience

1 1 Performance Optimization

Before publishing:

Check

Model Size

DAX Efficiency

Visual Count

Refresh Performance

Tools

Performance Analyzer.

1 **2** Documentation Standards

Why Documentation Matters

Recruiters often read documentation before reviewing dashboards.

Required Files

README.md

Project overview.

Data Dictionary

Field descriptions.

Business Requirements

Project objectives.

Screenshots

Dashboard previews.

1 **3** GitHub Portfolio Development

Why GitHub?

GitHub demonstrates professionalism and project organization.

Recommended Repository Structure

```
Project
├──
├── Dataset
├── PowerBI
├── Screenshots
```

- Documentation
- SQL
- README.md

Benefits

Visibility

Credibility

Portfolio Showcase

Power BI Portfolio Structure

Recommended Portfolio Categories

Sales Analytics

HR Analytics

Finance Analytics

Marketing Analytics

Supply Chain Analytics

Customer Analytics

Goal

Show versatility.

Case Study Presentation

Structure

Business Problem

Dataset

KPIs

Dashboard

Insights

Recommendations

Example

Revenue declined 15%
in the South region.

Recommendation:

Increase marketing investment.

Interview Preparation

Common Questions

Explain Star Schema.

Difference Between Measures and Columns?

Explain CALCULATE().

Explain Row Context vs Filter Context.

How Does RLS Work?

Explain Incremental Refresh.

Describe a Power BI Project.

Preparation Strategy

Build Projects

Document Decisions

Practice Presentations

Real-World Project Ideas

Project 1

Sales Analytics Dashboard

Project 2

HR Analytics Dashboard

Project 3

Financial Performance Dashboard

Project 4

Customer Segmentation Dashboard

Project 5

Supply Chain Analytics

Project 6

E-Commerce Analytics

Project 7

Marketing Campaign Analytics

Project 8

Call Center Analytics

Project 9

Healthcare Analytics

Project 10

Retail Performance Dashboard

Best Practices

Focus on Business Value

Use Realistic Datasets

Create Professional Documentation

Optimize Performance

Publish Screenshots

Explain Insights

Maintain Consistent Design

Common Mistakes

No Documentation

Poor Dashboard Design

Missing Business Context

Overcomplicated Visuals

Weak GitHub Structure

No KPI Definitions



Key Terminologies

Term	Definition
KPI	Key Performance Indicator
Stakeholder	Business user of reports
Portfolio	Collection of projects
Documentation	Project explanation files
Dashboard	Visual analytics solution
Case Study	Structured project presentation
README	Repository overview file
Data Dictionary	Dataset field definitions
Business Requirement	Project objective
Insight	Actionable finding from data



Practice Exercises

Exercise 1

Define KPIs for:

- Sales
 - HR
 - Finance
-

Exercise 2

Design a GitHub repository structure.

Exercise 3

Create a project documentation template.

Exercise 4

Prepare a dashboard wireframe.

Exercise 5

Present a project as a business case study.



Capstone Project

Executive Sales Analytics Solution

Business Problem

A retail company wants to improve sales performance.

Dataset

Sales Transactions

Customer Data

Product Data

Region Data

Requirements

Data Preparation

- Clean Data
 - Remove Duplicates
-

Data Modeling

- Star Schema
-

DAX

Create:

Revenue
Profit

Margin %
Growth %
YTD Revenue

Dashboard

Include:

- KPI Cards
- Revenue Trend
- Product Analysis
- Regional Analysis
- Customer Analysis

Optimization

- Performance Analyzer
- Incremental Refresh

Security

- Dynamic RLS

Documentation

- README
- Data Dictionary
- Business Requirements
- Screenshots

Deliverables

1. PBIX File
2. Documentation
3. GitHub Repository
4. Dashboard Screenshots
5. Business Insights Report



Module Summary

In this module, you learned:

- ✓ End-to-End Project Lifecycle

- ✓ Requirement Gathering
- ✓ KPI Identification
- ✓ Project Planning
- ✓ Dataset Selection
- ✓ Data Cleaning
- ✓ Data Modeling
- ✓ DAX Development
- ✓ Dashboard Design
- ✓ Performance Optimization
- ✓ Documentation Standards
- ✓ GitHub Portfolio Development
- ✓ Case Study Presentation
- ✓ Interview Preparation
- ✓ Real-World Project Ideas

Projects are the bridge between learning Power BI and becoming employable. A strong portfolio demonstrates technical skills, business understanding, communication ability, and problem-solving capability.

Next Module

Module 23: Power BI Interview Preparation & Certification Guide

Topics Covered:

- Power BI Interview Questions
- Scenario-Based Questions
- DAX Interview Questions
- Data Modeling Questions
- SQL for Power BI Interviews
- Power BI Certification Paths
- PL-300 Exam Preparation
- Resume Building
- Portfolio Review
- Career Roadmap